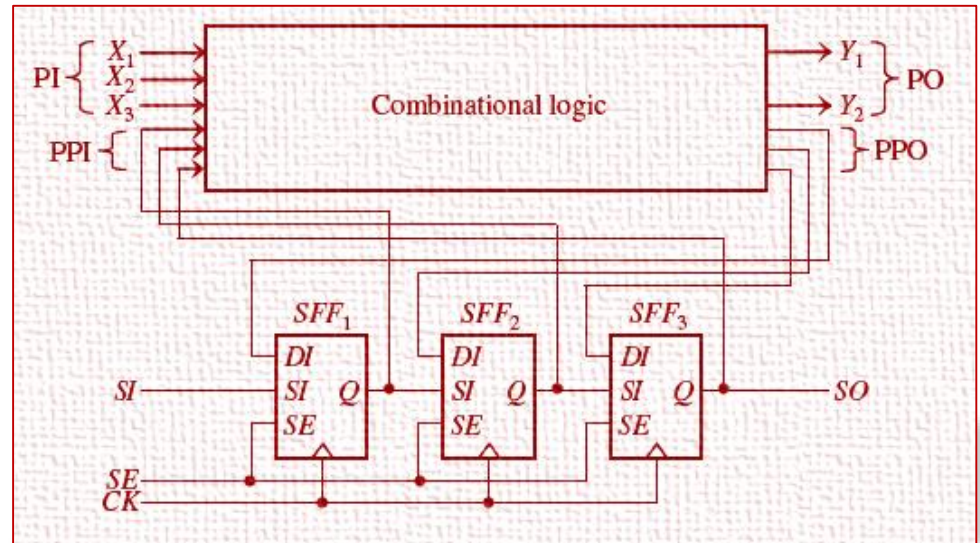


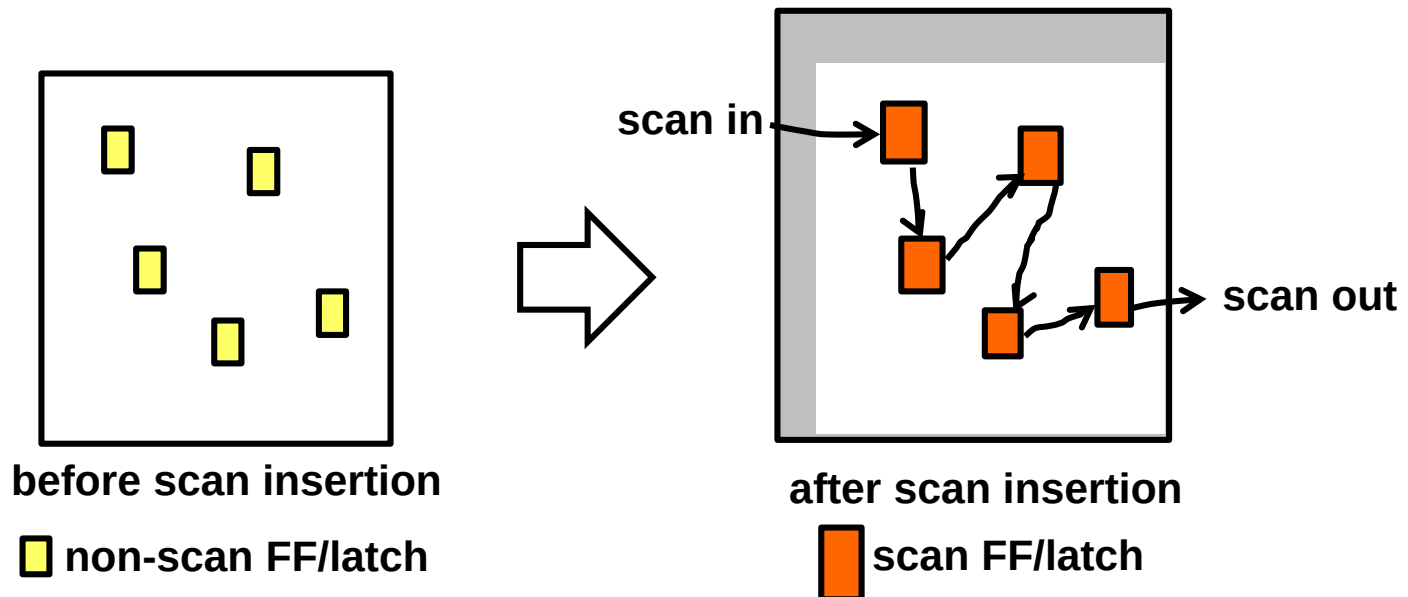
DFT - Part 1

- Introduction
- Internal Scan
 - ◆ FF-based
 - * MUXed-D scan (1973, Stanford)
 - * Clocked scan (1968, NEC)
 - ◆ Latch-based
 - * LSSD (1977, IBM)
- Scan Design Flow
- Issues and Solutions
- Conclusion



Scan Chains

- **Scan**: connect (internal) FF/latches as shift register
 - ♦ Control and observe FF/latches **in test mode**
 - ♦ Remain original function **in normal mode**
- Proposed in early 1970's [Williams 73][Eichelberger 77]
 - ♦ **Most important DFT** for synchronous digital circuits
- **Scan chain insertion**: aka. **DFT insertion, DFT synthesis**
 - ♦ 1. Replace FF/latch 2. Stitch FF/latch into a chain



Scan Flip-Flop (SFF)

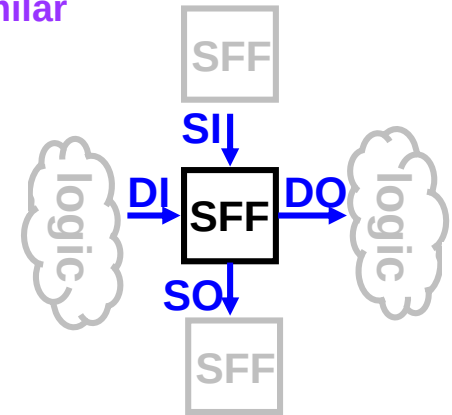
*scan latch is similar

- SFF has four main pins:

- ♦ Scan Chain: **Scan Input (SI), Scan output (SO)**

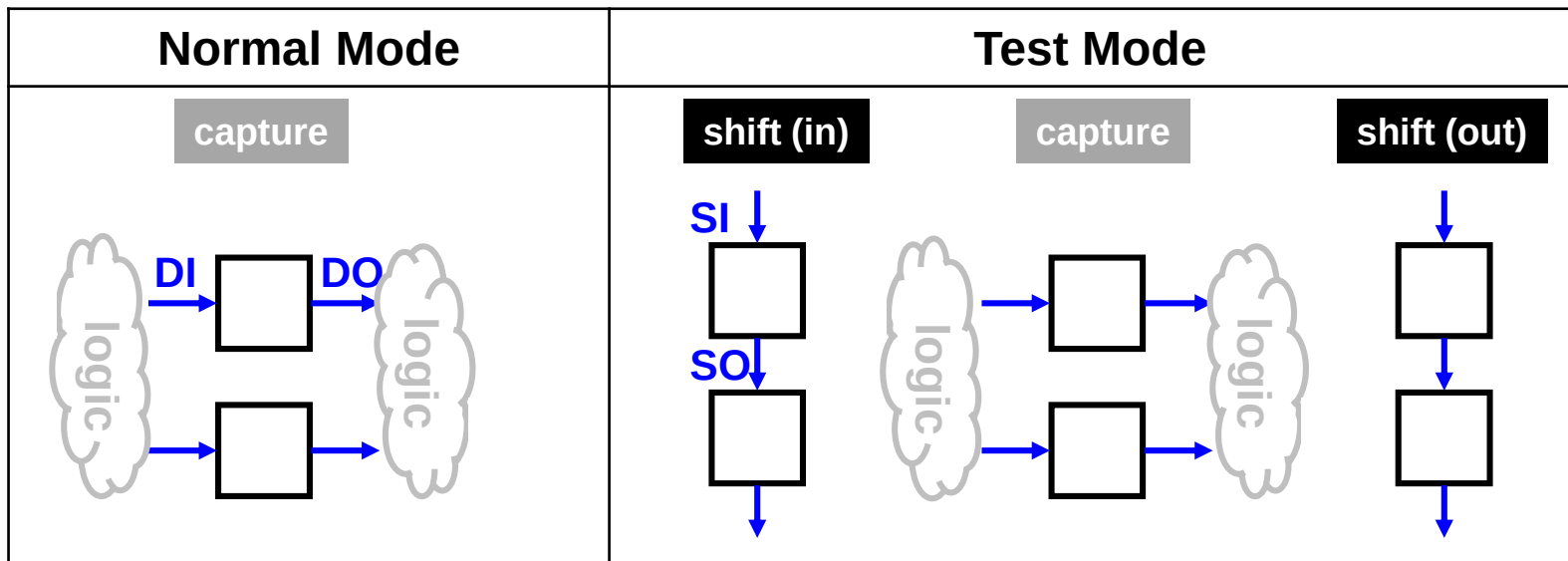
- ♦ Logic: **Data Input (DI), Data Output (DO)**

* DO and SO can be shared

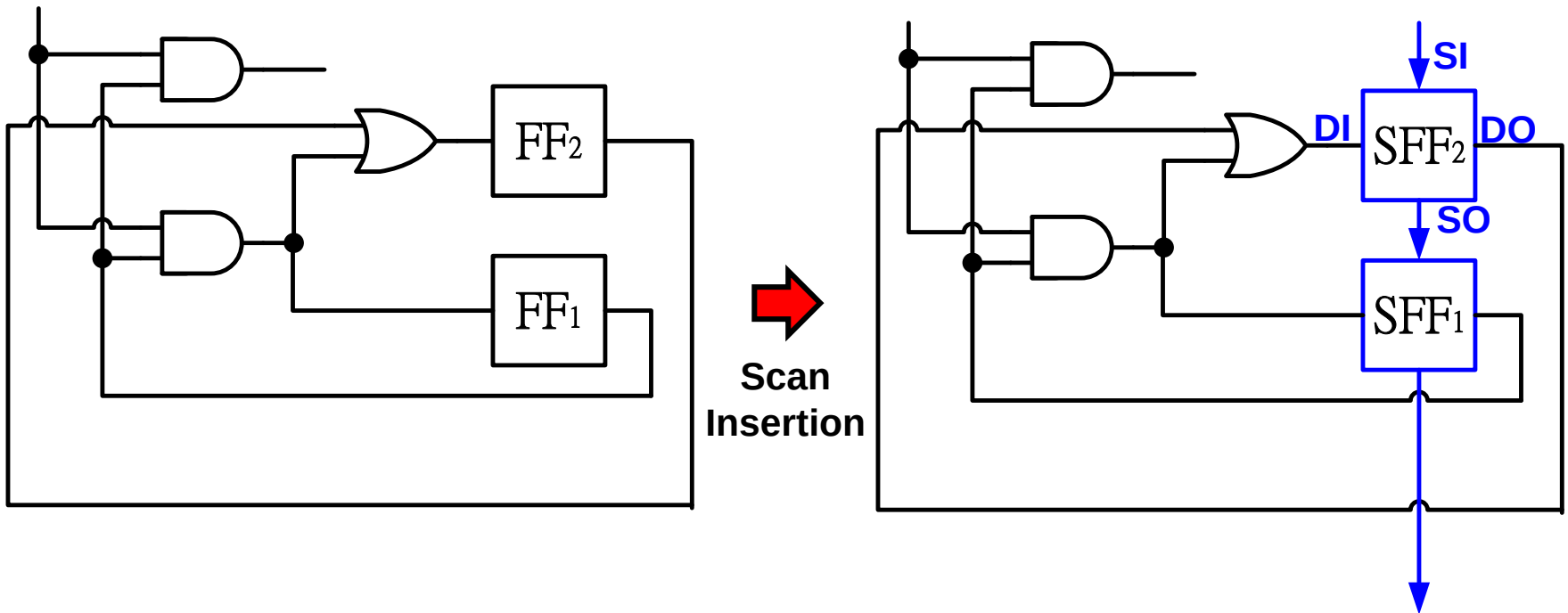


- SFF has two functions: **shift** and **capture**

- Circuit has two operation modes: **Normal** mode and **Test** mode



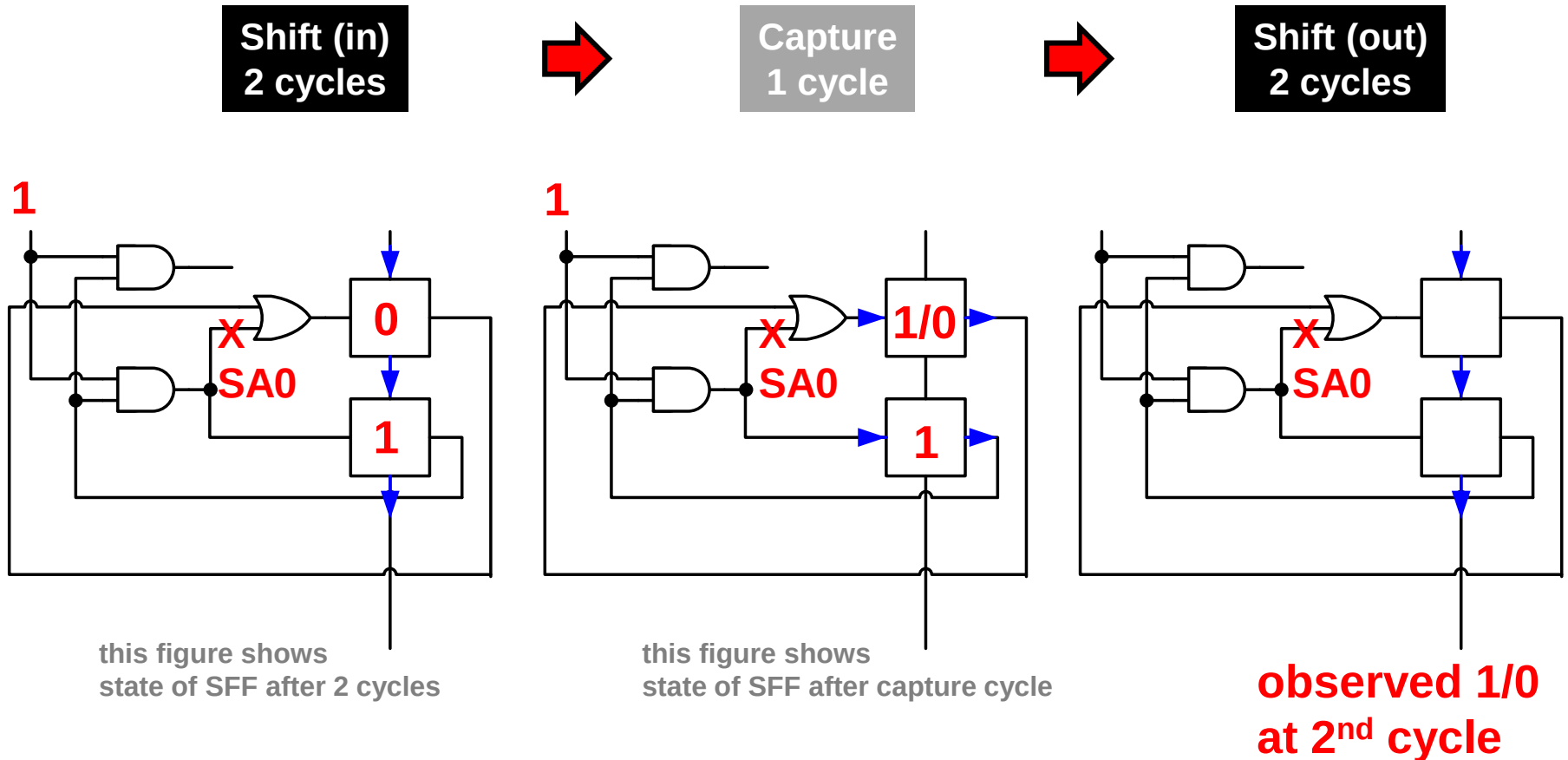
Example: Scan Insertion



- **Scan-FF same as non-scan FF**



Example: Test Mode



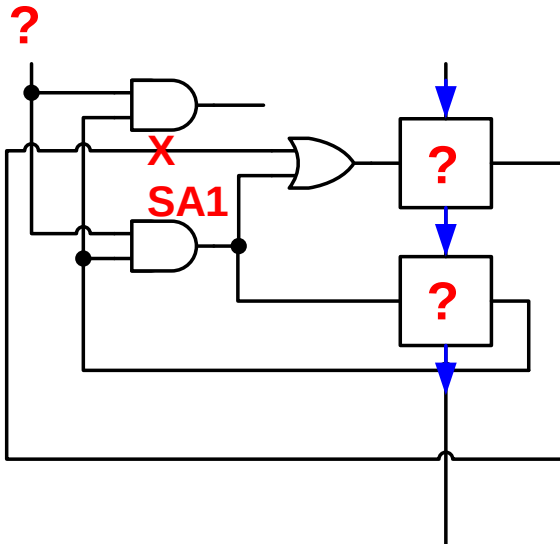
Fault Detected in Test Mode

NOTE: this fault untestable in Normal Mode. why?

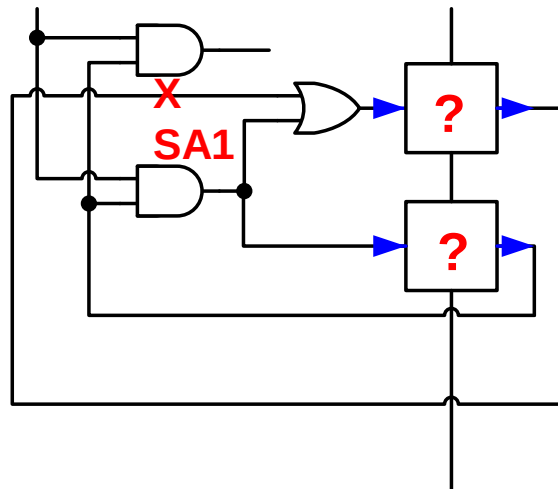
Quiz

Q: Consider SA1 fault, please fill in values of ?

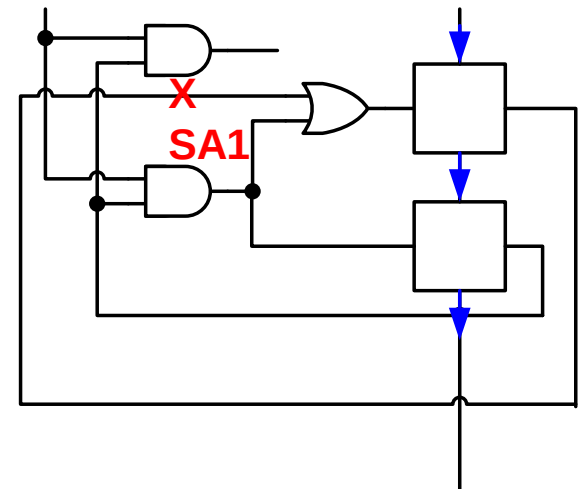
Shift (in)
after 2 cycles



Capture
after 1 cycle



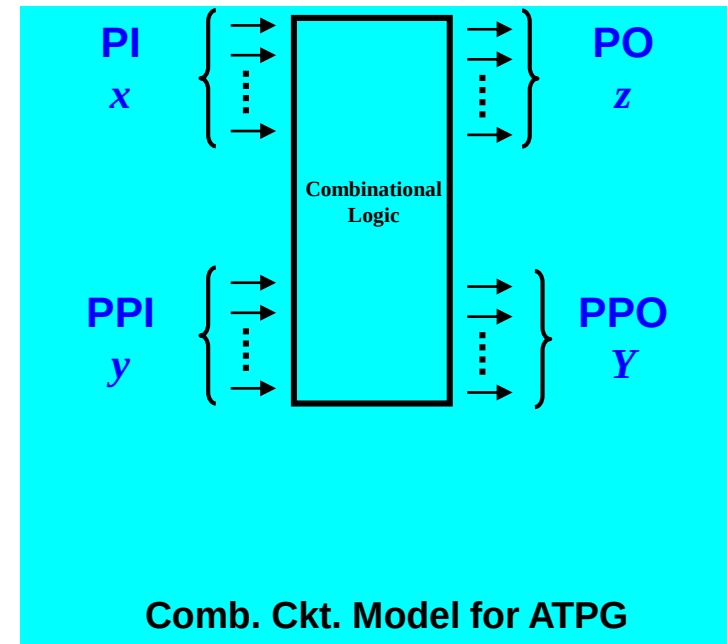
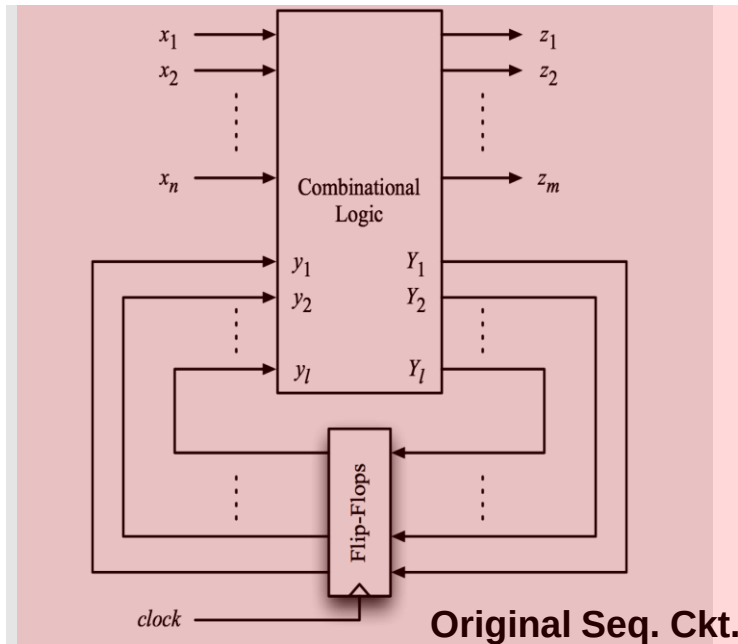
Shift (out)
after 2 cycles



observed ?
@ ? cycle

Scan Turns Seq. Ckt. to Comb. Ckt.

- Scan turns sequential ckt into combinational ckt in test mode
 - ♦ ATPG sees only **comb. ckt. model**
- SFF DO become **Pseudo Primary Input (PPI)**, fully controllable
- SFF DI become **Pseudo Primary Output (PPO)**, fully observable



Scan Make ATPG Easier!

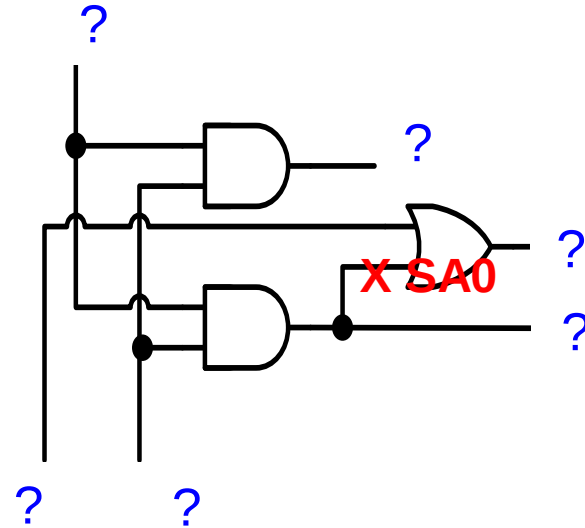
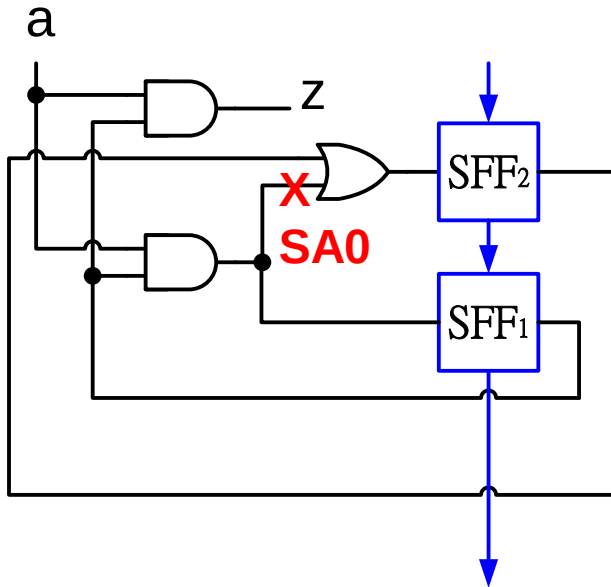
QUIZ

We insert scan into this circuit so we can remove FF in ATPG model.

Q1: Which pins are PI? PO? PPI? PPO?

Q2: If we want to detect SA0 fault, what are their values?

ANS:



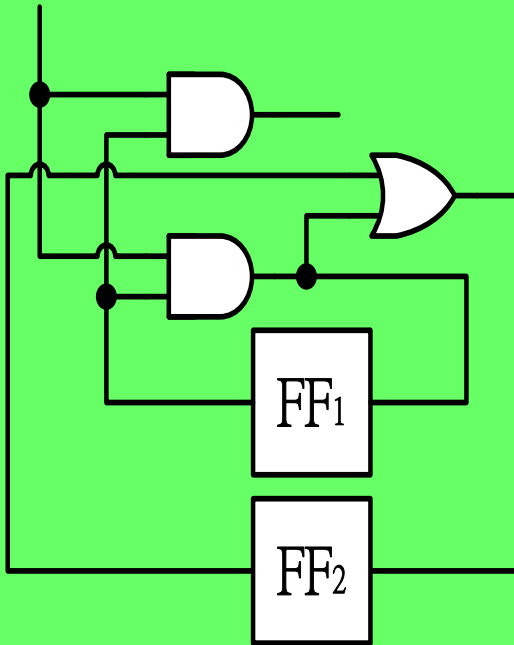
Pros and Cons of Scan

- Advantages of scan
 - ♦ **Systematic DFT**, not *ad hoc*. Many automatic tools.
 - ♦ **Easy ATPG**: faster run time and higher fault coverage (see 11.4)
 - ♦ Easy **silicon debugging** or *diagnosis*
- Disadvantages of scan
 - ♦ **Area overhead** (typically, 5~10% OH acceptable)
 - * larger SFF area + routing for scan chains
 - ♦ **Performance overhead**
 - * SFF has longer setup time, hold time
 - ♦ **Pin overhead**
 - * scan in, scan out, *scan enable*, *scan clocks* ... (11.3, 11.5)
 - ♦ **Power overhead**
 - ♦ **Extra design efforts**
 - * scan insertion, verification (11.7, 11.8)

QUIZ

We insert scan into this circuit and replace non-scan FF by scan FF.

Q: Given the gate area table, what is area overhead of scan DFT?



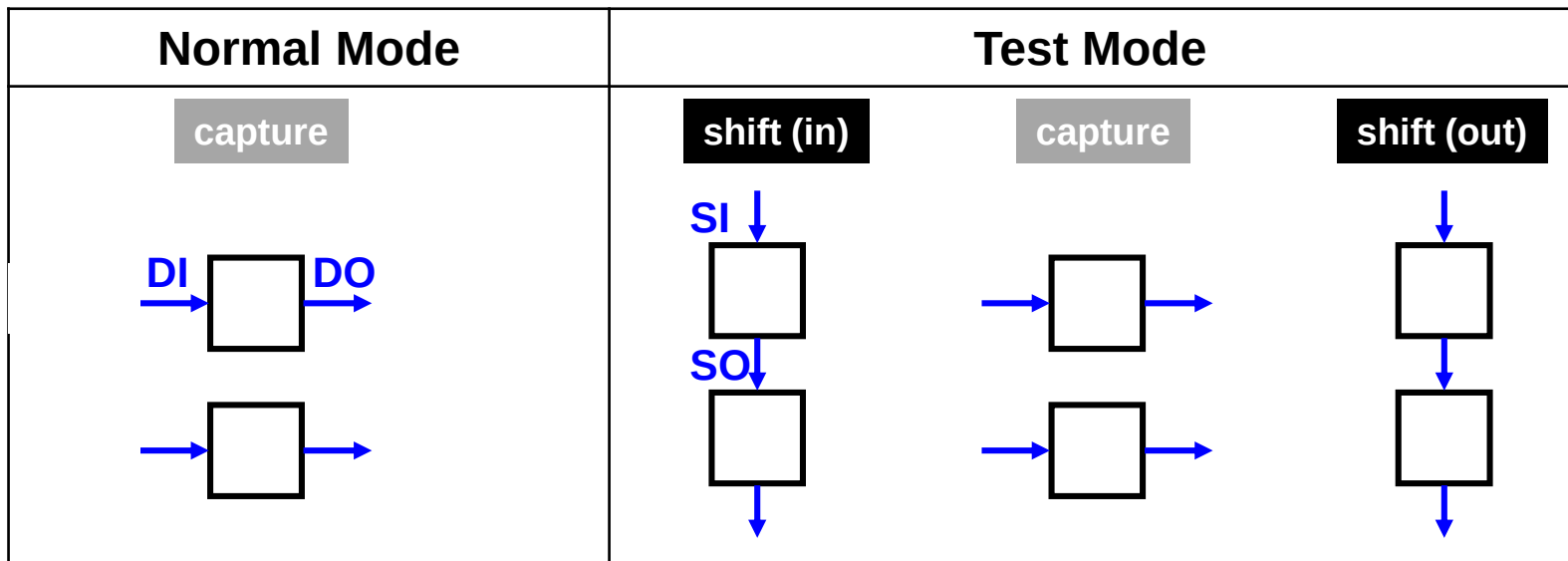
Gate	area
OR	3
AND	3
non-scan FF	5
scan FF	6

ANS:

$$\text{area O.H.} = \frac{\text{area}_{\text{afterDFT}} - \text{area}_{\text{beforeDFT}}}{\text{area}_{\text{beforeDFT}}} =$$

Summary

- Scan is **most popular DFT** for digital VLSI
- Scan FF has four pins: **SI, SO, DI, DO**
- Circuit has two operation modes: **normal** mode, **test** mode
 - ♦ Test mode: **shift** (in), **capture**, **shift** (out)
- Scan makes ATPG easier at cost of **overhead** (area/performance ...)



FFT

- Q1: This stuck-at zero fault is untestable in normal mode, why?
- Q2: Since it is untestable, why do we care about it?

